

RESEARCH ARTICLE

Evaluating the Integration of Spatial Features in Machine Learning for Model Accuracy and Residual Autocorrelation

Hyeongmo Koo¹  | Musang Yoo¹  | Hyeyun Kang¹  | Youngchul Cho¹  | Min Jeong^{1,2} 

¹Department of Geoinformatics, The University of Seoul, Seoul, Republic of Korea | ²School of Economic, Political, and Policy Sciences, The University of Texas at Dallas, Richardson, Texas, USA

Correspondence: Musang Yoo (yu1364@uos.ac.kr)

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ABSTRACT

Machine learning is increasingly applied in spatial data analysis, with spatial features offering potential improvements in its predictive accuracy. However, comparative evaluations of different spatial feature types remain limited, especially under varying relationships between input and response variables and differing levels of spatial autocorrelation. This study systematically evaluates the effectiveness of three spatial feature types, including spatial coordinates, spatial lags, and Moran eigenvector spatial filters (MESF), across four machine learning models with distinct estimation procedures. Using simulation experiments, the analysis examines both first-order effects (correlation between input and response variables) and second-order effects (spatial autocorrelation in the response variable). To validate the practical applicability of the simulation findings, an empirical analysis is subsequently conducted using municipal-level homeownership rate data from South Korea. Results from both the simulation and empirical analysis show that spatial features improve model accuracy more when first-order effects are weak. MESF consistently provides the greatest accuracy improvement, primarily due to its capacity to capture diverse spatial patterns through orthogonal eigenvectors. Under strong second-order effects, spatial features again enhance accuracy, with MESF proving most effective. MESF also best reduces residual spatial autocorrelation by modeling spatial dependence without confounding input-response relationships, whereas spatial lags often overcorrect, reversing autocorrelation direction. These findings highlight that the effectiveness of spatial features varies more by feature type than by model type. By quantifying their impact on accuracy and residual autocorrelation, this study provides empirical support for incorporating spatial features in machine learning, offering foundational insights for spatial machine learning development.

1 | Introduction

Machine learning is widely used in various analytical domains, including spatial analysis, due to its flexibility in applying models tailored to different analytical objectives. In the field of spatial analysis, deep learning is prominently employed in remote sensing tasks such as object detection and image classification (Bui et al. 2024; Lee et al. 2023). Beyond remote sensing, a range of models, including random forests and boosting techniques, are also actively utilized in diverse areas such as urban

studies and epidemiology (Jun 2021; Lombard et al. 2021; Mehta et al. 2020).

Spatial data exhibits inherent properties such as spatial autocorrelation, which complicates the assumption of stationarity in modeling (Miller 2012). Models that fail to adequately represent spatial structures violate this assumption, resulting in spatially structured systematic errors, reduced predictive accuracy (Hu et al. 2022; Song and Kim 2023), and inflated cross-validation estimates (Kattenborn et al. 2022; Yoo and Koo 2025).

To mitigate these issues, recent methodological advances have focused on strategies to explicitly incorporate spatial dependence into regression based machine learning models (Kopczewska 2022; Nikparvar and Thill 2021). These strategies can be broadly categorized into two approaches: model-specific adaptations and feature-based methods. The former embeds spatial structure directly into the estimation procedure, as seen in geographically weighted or spatially adaptive extensions (e.g., geographical random forests) (Georganos et al. 2021). The latter encodes spatial structures as additional input variables (e.g., spatial coordinates, spatial lags, or Moran eigenvector spatial filters [MESF]) allowing for integration into a wide range of algorithms without modifying estimation procedures (Islam et al. 2022; Li and Peng 2025). While recent studies have also explored integrating spatial features into deep learning architectures to address computational costs and non-stationarity (Wang et al. 2025), feature-based approaches in standard machine learning models offer distinct advantages in terms of flexibility and broad applicability (Barry et al. 2024; Jamal et al. 2023).

Although the utility of spatial features in machine learning for spatial data analysis is increasingly acknowledged (Kopczewska 2022; Nikparvar and Thill 2021), comparative studies evaluating their effectiveness across different spatial feature types remain limited, particularly under varying relationships between response and input variables and differing levels of spatial autocorrelation in the response variable. For example, Liu et al. (2022) assessed whether spatial lags and MESF enhance model accuracy using random forests, LASSO, and geographically weighted regression (GWR). Similarly, Hu et al. (2022) examined the contribution of spatial coordinates, local Moran statistics, and MESF to the predictive performance of linear regression and random forest models. Koo et al. (2024) also analyzed the effects of spatial feature incorporation on prediction accuracy and compared the utility of various spatial feature types and machine learning models. However, these studies primarily investigate spatial feature effectiveness under fixed conditions characterized by strong correlations between input and response variables and high levels of positive spatial autocorrelation. Consequently, the comparative performance of these features in more complex scenarios (e.g., under weak first-order effects and negative spatial autocorrelation) remains a critical gap in the literature.

This study investigates the effectiveness of spatial features in regression under diverse conditions characterized by differing degrees of correlation between input and response variables, that is, first-order effects, and varying levels of spatial autocorrelation in response variables, that is, second-order effects. Specifically, the analysis employs simulation experiments focused on bivariate regression problems with a single input variable. This controlled setting is designed to quantify the marginal contribution of spatial features while isolating them from the confounding effects of complex multivariate interactions. To validate the practical applicability of the simulation findings, the study further conducts an empirical analysis predicting municipal-level homeownership rates in South Korea. By applying the models to a real-world dataset exhibiting strong spatial autocorrelation and involving multiple covariates, we examine whether the comparative advantages of specific spatial features persist in practical modeling tasks. Ultimately, this study is expected to provide quantitative evidence of the advantages of integrating spatial features in machine learning models, thereby establishing foundational insights to inform future developments in spatial machine learning.

2 | Method

2.1 | Analysis Procedures

This section presents the simulation procedure to evaluate the effects of spatial feature integration in machine learning models (Figure 1). The simulation analysis begins with a baseline experiment that evaluates model performance without spatial features, focusing on the influence of first-order effects (i.e., correlation between input and response variables) and second-order effects (i.e., spatial autocorrelation in the response variable) on both prediction accuracy and the spatial autocorrelation in model residuals. This baseline serves to identify the isolated impact of the spatial characteristics on model outcomes. Subsequently, a second simulation experiment examines the extent to which incorporating spatial features improves model accuracy under varying levels of first- and second-order effects. Then, it investigates whether the inclusion of spatial features mitigates spatial autocorrelation in model residuals. Finally, the study extends the evaluation through an empirical analysis

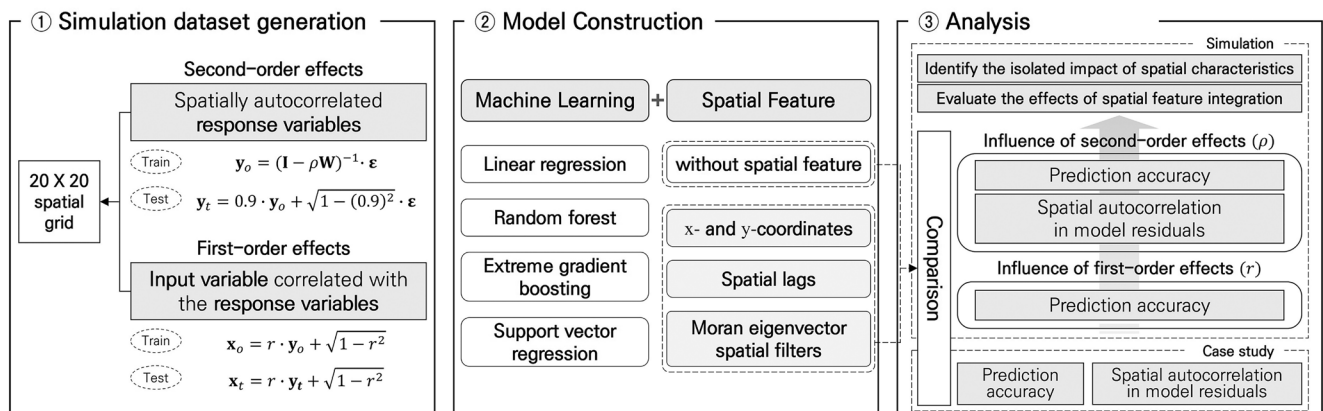


FIGURE 1 | The procedure of the simulation analyses.

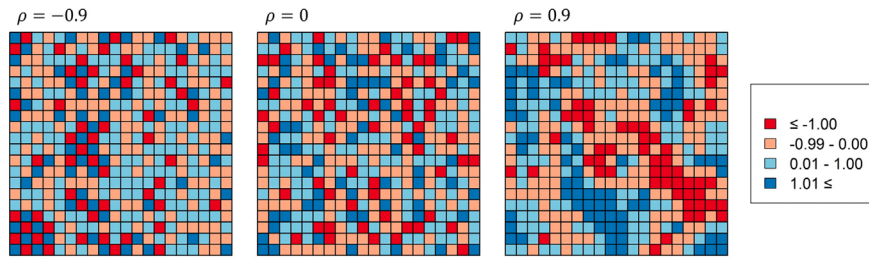


FIGURE 2 | The examples of the randomly generated spatially autocorrelated response variable with varying spatial autocorrelation.

using municipal-level homeownership rate data from South Korea. This practical application serves to validate the simulation findings, examining whether the comparative advantages of spatial features regarding accuracy and residual mitigation persist in a real-world multivariate setting.

The following describes the procedure for generating the simulation dataset. A total of 1000 synthetic training datasets and 1000 corresponding testing datasets are generated, each constructed on a 20-by-20 spatial grid. To generate the training dataset, the response variable is first created by incorporating second-order effects, specifically varying levels of spatial autocorrelation. This is achieved using a spatial autoregressive (SAR) process (Anselin 1995; Koo et al. 2019) as follows:

$$\mathbf{y}_o = (\mathbf{I} - \rho \mathbf{W})^{-1} \cdot \boldsymbol{\varepsilon}$$

where $\mathbf{y}_o$ represents the spatially autocorrelated response variables; $\mathbf{I}$ is an n -by- n identity matrix; $\boldsymbol{\varepsilon}$ is a random vector drawn from a standard normal distribution; and $\mathbf{W}$ is a row-standardized spatial weights matrix. $\mathbf{W}$ defines neighbors based on queen contiguity weights. ρ is used to adjust the level of spatial autocorrelation in a response variable, which is randomly drawn from a uniform distribution ranging between -1 and 1 (Figure 2).

Subsequently, a random input variable in the training dataset is generated to reflect first-order effects by introducing varying levels of correlation with the response variable. The correlated input variable is created using Cholesky decomposition as follows (Yoo and Koo 2025):

$$\mathbf{x}_o = r \cdot \mathbf{y}_o + \sqrt{1 - r^2} \cdot \boldsymbol{\varepsilon}$$

where $\mathbf{x}_o$ is the input variable correlated with the response variable ($\mathbf{y}_o$). The magnitude of the first-order effect is determined by the Pearson's correlation coefficient (r), which is randomly also drawn from a uniform distribution ranging from -1 to 1 .

The process for constructing the testing dataset is as follows. The testing response variable ($\mathbf{y}_t$) is first generated to exhibit a strong positive correlation with the response variable ($\mathbf{y}_o$) from the training dataset. This is also achieved using the same decomposition method as follows:

$$\mathbf{y}_t = 0.9 \cdot \mathbf{y}_o + \sqrt{1 - (0.9)^2} \cdot \boldsymbol{\varepsilon}$$

The correlation coefficient between the two response variables is fixed at 0.9 , which serves as the target model accuracy when

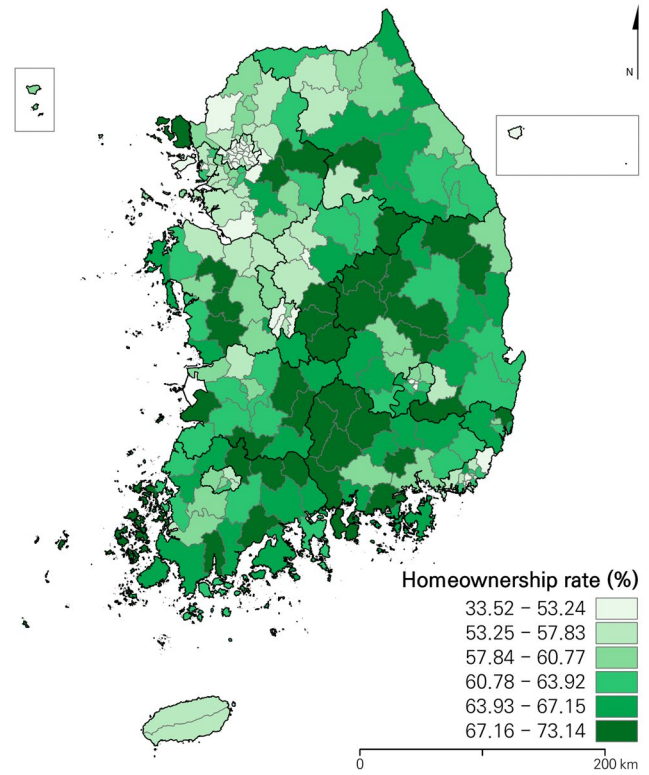


FIGURE 3 | Homeownership rate in South Korea, 2024.

comparing training and testing datasets. Subsequently, to ensure a consistent level of correlation with the response variable ($\mathbf{y}_t$), an input variable in the testing dataset ($\mathbf{x}_t$) is generated using the same r value used in the training dataset generation process as follows:

$$\mathbf{x}_t = r \cdot \mathbf{y}_t + \sqrt{1 - r^2} \cdot \boldsymbol{\varepsilon}$$

For the empirical evaluation, the study utilizes municipal-level data from South Korea to predict the homeownership rate (HR) (Figure 3). The response variable, HR, exhibits strong positive spatial autocorrelation in both the training (2019) and test (2024) datasets, with Moran coefficients of 0.556 ($z = 13.95$) and 0.582 ($z = 14.59$), respectively. Two input variables are incorporated: the proportion of household heads aged 65 or older (HA65+) and the official land price (LP). Data for HR and HA65+ were derived from the Housing Ownership Statistics and Population Census by Statistics Korea, while LP is obtained from the National Statistical Map provided by the National Geographic Information Institute. The models are trained using 2019 data

and tested with 2024 data. Despite the temporal gap, the correlations between the input and response variables remain consistent (approximately 0.6 for HA65+ and -0.6 for LP), and the spatial structure of the response variable remains stable. These consistent characteristics support the suitability of this dataset for validating the effectiveness of spatial features in real-world modeling scenarios.

2.2 | Spatial Features

This section begins by describing the types of spatial features. The methods for generating spatial features can be broadly categorized into two approaches (Koo et al. 2024; Nikparvar and Thill 2021). The first approach involves using spatial reference information, while the second approach focuses on incorporating attributes of spatial objects or relational information between objects.

The first approach, using spatial reference information, is exemplified by incorporating spatial coordinates (e.g., latitude-longitude or planar coordinates) as input variables (Behrens et al. 2018; Cracknell and Reading 2014). Additionally, spatial structure of a study area can be classified and then encoded using nominal scales (Anselin and Arribas-Bel 2013; Chun and Griffith 2011; Tiefelsdorf and Boots 1995). Eigenvector spatial filtering is also a widely used method for representing spatial structure as input features (Islam et al. 2022). When the response variable is spatially distributed, it often exhibits spatial structure, which can enhance the predictive accuracy of machine learning models. However, careful consideration is required, as such spatial features—particularly spatial coordinates—tend to exhibit high spatial autocorrelation, which can undermine the independence between variables (Griffith et al. 2017). This may result in reduced predictive accuracy and negative effects during the learning process in machine learning models (Setiyorini et al. 2017).

The second approach involves using attributes of spatial objects or relationships between objects as input variables. First, this approach uses typical attributes of spatial objects including elevation, slope, and area of the spatial objects (Hong et al. 2017; Li et al. 2024). Moreover, relational information between spatial objects, such as the average values of neighboring areas, that is, spatial lags, are also employed (Kiely and Bastian 2020), which are generally suitable when strong positive spatial autocorrelation exists. Furthermore, local indicators of spatial association (e.g., local Moran coefficients) are often used to capture local variation with respect to neighboring regions (Ghimire et al. 2010; Hu et al. 2022).

This study selects three types of spatial features for evaluating their effectiveness, which include x - and y -coordinates, spatial lags, and MESF. First, the x - and y -coordinates are represented through first-, second-, and third-order polynomial transformations, denoted as $S_{\text{coord}} = \{x, y, x^2, y^2, x^3, y^3\}$ (Chorley and Haggett 1965). This specification allows the model to account for nonlinear marginal spatial trends while reducing the risk of multicollinearity associated with interaction terms (e.g., xy). Subsequently, a feature selection method was applied to retain only those variables that significantly contributed to improving

model accuracy. Specific details regarding this feature selection procedure are provided in Section 2.3.

The following describes the procedure for generating spatial lag variables. Specifically, spatial lags represent the spatially weighted average of neighboring response values. This study defines neighbors based on the queen contiguity criterion, extending to higher orders (first-, second-, and third-order) (Anselin and Smirnov 1996). Consequently, the set of spatial lag features is formally expressed as $S_{\text{lag}} = \{W_1y, W_2y, W_3y\}$, where W_o represents the row-standardized spatial weights matrix corresponding to the o -th order neighbors. This formulation captures the spatial dependence structure at varying scales.

Lastly, spatial features were generated using MESF, which is originally developed to address the violation of the independence assumption of regression residuals in spatial data analyses (Griffith 2003). Specifically, MESF involves extracting eigenvectors from the following:

$$(\mathbf{I} - \mathbf{1}\mathbf{1}^T/n)\mathbf{C}(\mathbf{I} - \mathbf{1}\mathbf{1}^T/n)$$

where $\mathbf{I}$ is the identity matrix, $\mathbf{1}$ is a vector of ones, and T denotes its transpose. $\mathbf{C}$ represents a binary spatial weights matrix. Neighbors are defined using the queen contiguity criterion, consistent with the definition used for spatial lag variables. The eigenvectors extracted through this process represent a range of spatial structures, from patterns of strong positive to strong negative spatial autocorrelation across the study area (Tiefelsdorf and Boots 1995). Moreover, since the eigenvectors are orthogonal to each other, they can be used as independent variables in regression models.

Utilizing MESF as spatial features requires an additional selection procedure. Prior to applying feature selection methods, a candidate set of eigenvector spatial filters is constructed to reduce model complexity (Griffith 2003), a step that is also a critical determinant of MESF performance (Chun et al. 2016). To capture both positive and negative spatial autocorrelation mixed patterns (Hu et al. 2022), this study selects only those eigenvectors whose corresponding eigenvalues are either at least 15% greater than the largest eigenvalue or at least 15% smaller than the smallest eigenvalue.

2.3 | Machine Learning Models and Hyperparameter Tuning

This study explores the effectiveness of spatial features using several widely adopted machine learning models with distinct estimation procedures and varying levels of flexibility, including linear regression (LM), boosting (BT), random forests (RF), and support vector regression (SVR) (James et al. 2013). First, LM has traditionally been used to explore linear relationships between input and response variables, and in this study, it serves as the baseline model.

This study employs RF, which is an ensemble learning method built on decision trees. It builds multiple decision trees by randomly selecting subsets of input variables at each step and then aggregates them into a single model (Breiman 2001). In

the baseline model (without spatial features), which relies on a single input variable, this random feature selection process becomes redundant. Consequently, the algorithm operates similarly to bagging. However, when spatial features are included as additional input variables, the RF algorithm employs its standard mechanism of selecting random subsets of variables at each node split. For the RF, the number of trees is set to 500. The number of randomly selected variables at each split in the models with spatial features is determined to minimize the root mean squared error (RMSE) based on the Out-of-Bag (OOB) samples. The tuning procedures are independently performed in each of the 1000 simulation iterations.

This analysis uses BT, specifically extreme gradient boosting (XGBoost) based on decision trees (Chen et al. 2015), which generally constructs trees sequentially, reducing model error at each step according to a specified learning rate. In this study, the maximum tree depth and learning rate are fixed at 2 and 0.05, respectively. The number of boosting rounds (i.e., trees) varies from 1 to 500 in increments of 10, and the optimal hyperparameters are selected based on 5-fold cross-validation using RMSE.

This study utilizes SVR. One key advantage of SVRs is their ability to model nonlinear decision boundaries using kernel functions; therefore, this analysis adopts a radial basis function (RBF) kernel (Karatzoglou et al. 2004). The main hyperparameters of the RBF-based SVR model include the cost parameter, which controls the penalty for training errors, and the kernel parameter, which defines the radius of influence in the RBF kernel. These hyperparameters are tuned through 5-fold cross-validation.

Only statistically significant input variables, including spatial features, are retained using model-specific feature selection methods. For the LM, stepwise regression employing forward selection is utilized, whereby input variables are retained only if their inclusion yields a substantial reduction in the Akaike information criterion (AIC). In the case of RF, BT, and SVR models, recursive feature elimination (RFE) is implemented (Guyon et al. 2002). This technique systematically eliminates input variables that exhibit minimal contribution to the prediction of the response variable based on 5-fold cross-validation.

3 | Results

This study evaluates the effectiveness of spatial features through simulation experiments under varying degrees of first- and second-order effects, followed by an empirical validation using real-world data. First, the analysis examines model performance in the absence of spatial features. Specifically, model prediction accuracy is assessed using RMSE across different levels of correlation between the response and input variables. To facilitate clear visualization and interpretation, correlation levels were aggregated into 0.2 intervals. Preliminary analysis confirmed that this aggregation preserves the underlying trends observed at finer resolutions (e.g., 0.1 intervals). Subsequently, the impact of spatial autocorrelation in the response variable, measured using the z-values of Moran coefficients, is compared to that of the model residuals using the same statistic. Next, the study assesses the impact of incorporating spatial features into the models. Changes in RMSE following the inclusion of spatial features are examined across varying levels of first-order effects and compared by types of model and spatial feature. Additionally, the z-values of Moran coefficients for the model residuals are analyzed across different levels of spatial autocorrelation in the response variable, allowing for an inference regarding the effectiveness of spatial features in reducing residual spatial dependence. All evaluations are conducted using the testing dataset. Finally, the section concludes with the results of the empirical analysis, verifying whether the comparative advantages of spatial features regarding accuracy and residual mitigation persist in a practical multivariate regression problem.

3.1 | Impacts of First- and Second-Order Effects on Models Without Spatial Features

This section reports the evaluation results of model performance in the absence of spatial features. First, higher model accuracy under strong correlation between response and input variables suggests that the simulation process is appropriately capturing the underlying relationships (Figure 4a). Specifically, when the correlation level exceeds 0.7, regardless of whether it is positive or negative, the model RMSE is approximately 0.6 across models. In contrast, when the correlation level falls below 0.3, the

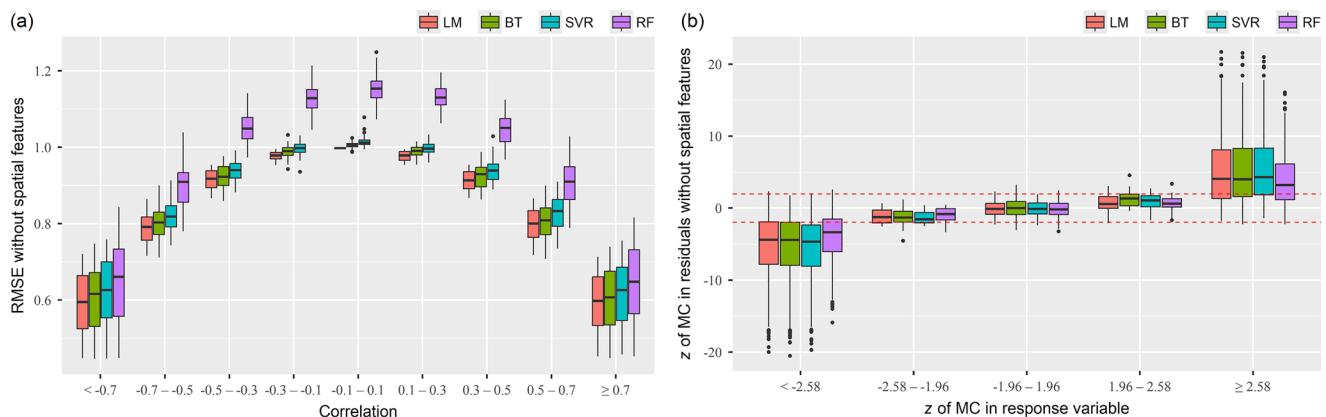


FIGURE 4 | Models without spatial features; (a) Relationship between RMSE and correlation levels between input and response variables (b) Relationship between the z-values of Moran coefficients in model residuals and those in response variables.

RMSE approaches 1.0. Given that the standard deviation of the response variable is fixed at 1.0, an RMSE close to 1.0 indicates little improvement over a naïve mean-based model, suggesting that the model accounts for a negligible proportion of the variance in the response variable.

Although differences in model accuracy exist among the models, they are not statistically significant based on *t*-tests. Since this simulation process is constructed based on a simple linear relationship between the input and response variables, the LM model consistently yields the lowest RMSE across all ranges of correlation. Specifically, when the correlation value is 0.8, the average RMSE is below 0.6, and when the correlation is 0.6, the average RMSE is approximately 0.8. In contrast, RF shows the lowest prediction accuracy (i.e., the highest RMSE) across all ranges of correlation. This outcome is likely due to underfitting, as the model includes only a single input variable, which limits its ability to capture complex interactions among predictors that typically enhance performance (Breiman et al. 1984; Ghogh and Crowley 2023). Moreover, when the correlation falls below 0.5, the RMSE exceeds 1.0. This threshold signifies that the model performs worse than a mean-only baseline, indicating a failure to capture the relationship between the input and response variables.

The analysis examining the effect of spatial autocorrelation in the response variable on that of the model residuals reveals that when the response variable exhibits strong spatial autocorrelation, the residuals also tend to display a similar level of spatial autocorrelation (Figure 4b). Specifically, in cases where the *z*-value of the Moran coefficient for the response variable exceeds 2.58, indicating significance at the 1% level, the residuals show statistically significant spatial autocorrelation at the 5% level, with *z*-values greater than 1.96 (red dotted lines in Figure 4b). The direction of spatial autocorrelation in the residuals corresponds to that of the response variable. That is, positive spatial autocorrelation in the response variable leads to positively autocorrelated residuals, while negative spatial autocorrelation in the response results in residuals with negative spatial dependence. Moreover, when the correlation between the response and input variables is strong, spatial autocorrelation in the residuals tends to diminish (refer to Figure A1 in Appendix A). In contrast, when this correlation is weak, the persistence of spatial structure in the residuals may suggest that the spatial pattern in the response variable is not adequately explained by the input variable, or that important explanatory variables are omitted (Ploton et al. 2020).

While the level of residual spatial autocorrelation differs slightly by model, the variations are not statistically significant (Figure 4b). In particular, LM, BT, and SVR produce largely similar results. RF shows a slight deviation, as the absolute values of the *z*-values of Moran coefficients in the residuals are generally lower than those observed in the other models. Nonetheless, this does not alter the overall interpretation: in cases where the *z*-value of the response variable exceeds 2.58, indicating significance at the 1% level, the residuals for all models, including RF, consistently show significant spatial autocorrelation at the 5% level, with corresponding *z*-values above 1.96.

3.2 | Impacts of Incorporating Spatial Features Under Varying First- and Second-Order Effects

This section examines the effect of incorporating spatial features into the models. First, the analysis examines the impact of incorporating spatial features on model accuracy by the degree of the first-order effect (Figure 5). This result generally indicates that when the first-order effect is strong, which means the input variable already accounts for a substantial portion of the variation in the response variable, the additional contribution of spatial features to improving model accuracy is limited. Specifically, across all models and spatial feature types, the reduction in RMSE becomes nearly negligible when the absolute value of the correlation between the input and response variables exceeds 0.7.

Incorporating spatial features generally yields greater improvements in model accuracy under weaker first-order effects, with MESF providing the most substantial improvement, followed by spatial lags and *x*- and *y*-coordinates (Figure 5). Specifically, when the absolute correlation between the input and response variables ranges from 0.1 to 0.3, MESF reduces RMSE by approximately 0.3. In contrast, spatial lags result in a reduction of less than 0.1, while *x*- and *y*-coordinates contribute minimally. These results indicate that under weak first-order effects, spatial features, particularly, MESF and spatial lags, enhance model accuracy by capturing the spatial structure in the response variable. Notably, MESF demonstrates the strongest capacity to explain spatial structure, likely due to its inclusion of a greater number of spatial features that better capture diverse spatial patterns. This finding differs from earlier findings (Koo et al. 2024) that identified spatial lags as the most effective.

The effect of incorporating spatial features on model accuracy varies across models, although it is generally smaller than the effect attributable to the type of spatial feature (Figure 5). Among the four models, the improvement in predictive accuracy is most pronounced in RF. Specifically, at correlation levels between 0.3 and 0.5, RF shows an RMSE reduction exceeding 0.4 with MESF. Even with spatial lags, the reduction reaches around 0.25, and notably, *x*- and *y*-coordinates, which show minimal effects in other models, yield a reduction close to 0.2. However, this outcome is largely attributable to underfitting, as RF in this simulation utilizes only a single input variable (Figure 4a). The structural sensitivity of RF to interactions among input variables amplifies the impact of spatial feature incorporation (Georganos et al. 2021), leading to a relatively greater improvement in model accuracy compared to the other models.

Next, this analysis investigates the impact of incorporating spatial features on model accuracy across different levels of the second-order effect, quantified by the *z*-values of Moran coefficients for the response variable (Figure 6). Incorporating spatial features also enhances model accuracy when the response variable exhibits second-order effects, with the degrees of improvement varying considerably across types of spatial features. Among the spatial features, MESF consistently results in the greatest improvement in model accuracy, followed by spatial

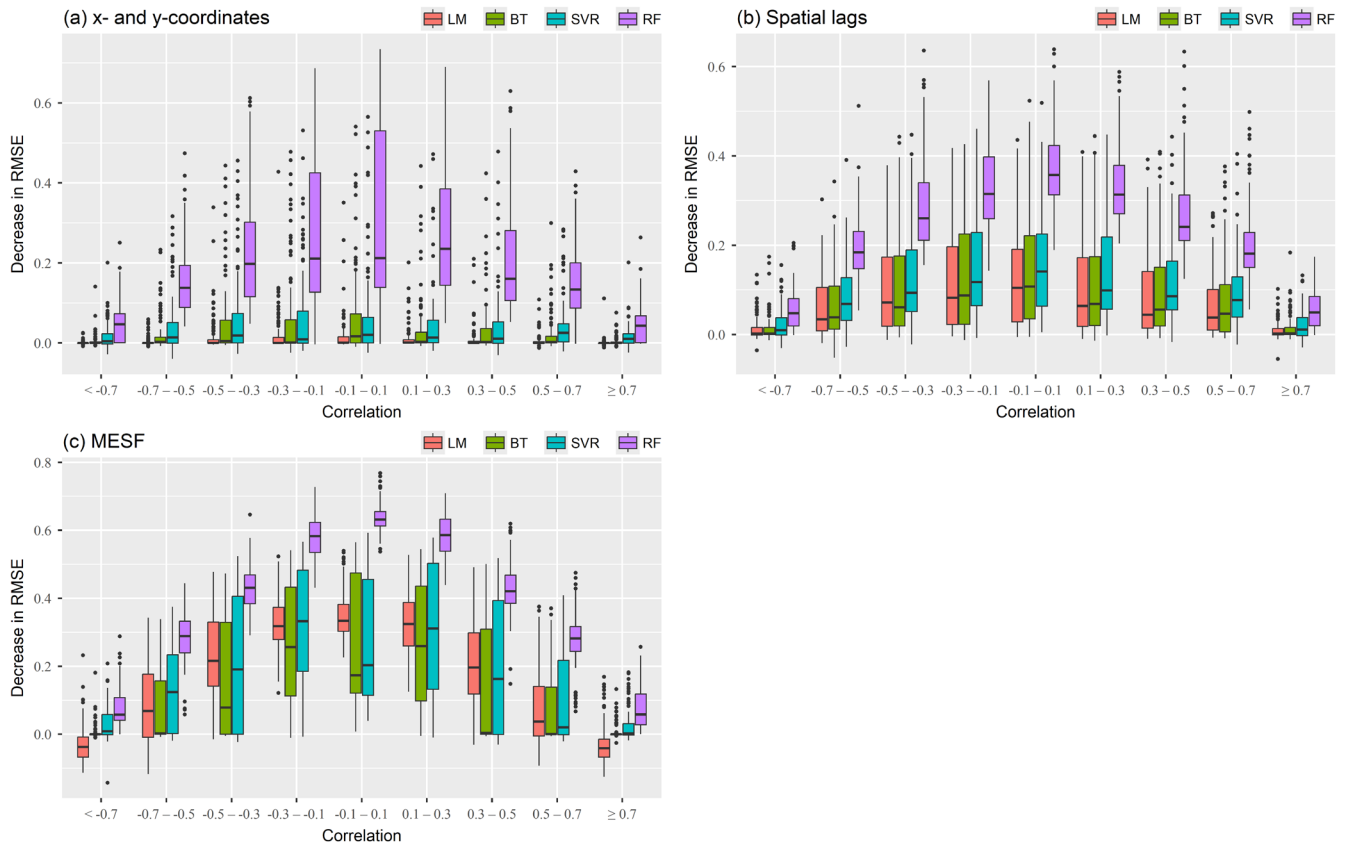


FIGURE 5 | RMSE reduction by incorporating spatial features across correlation levels.

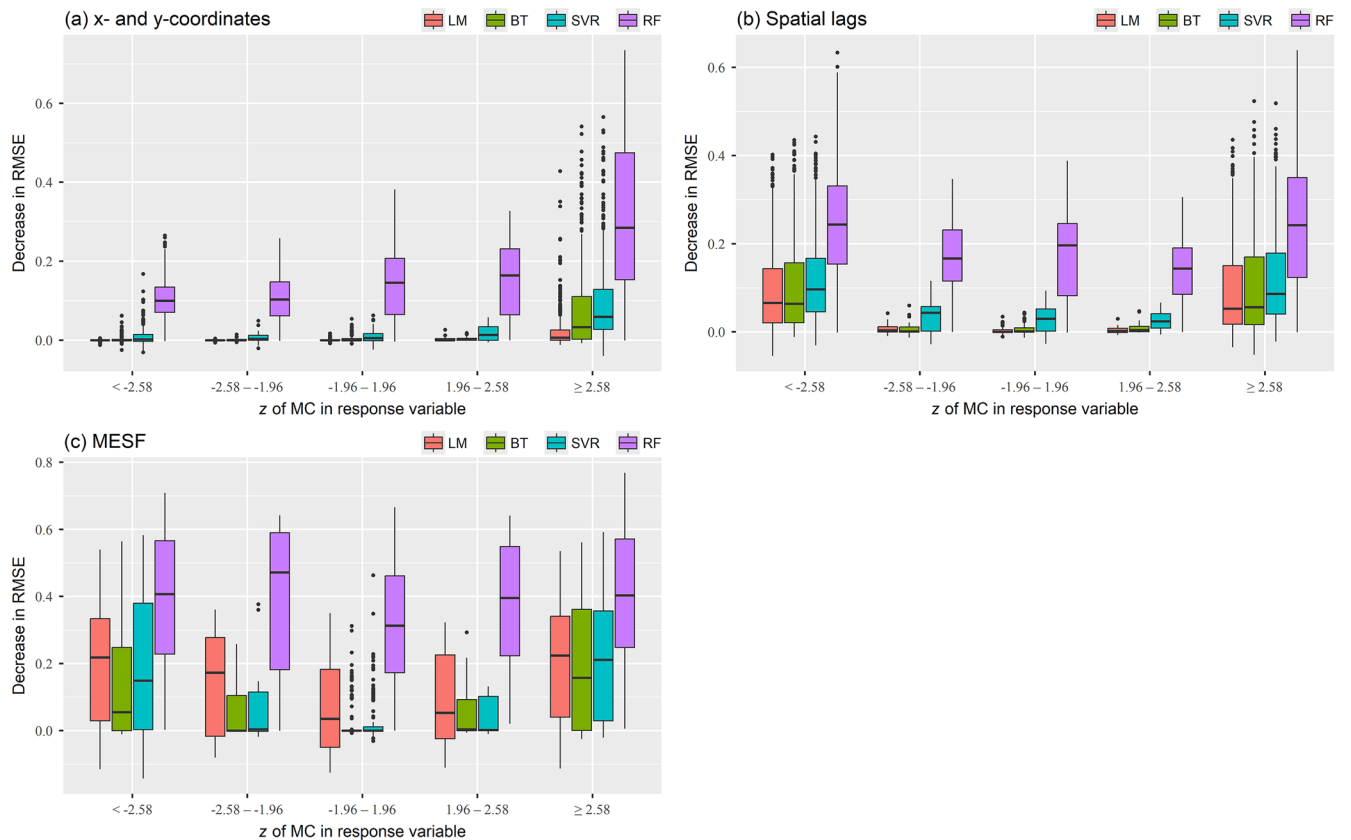


FIGURE 6 | RMSE reduction by incorporating spatial features across z -values of Moran coefficients in response variables.

lags and x- and y-coordinates. Specifically, when the absolute z-value of the Moran coefficient in the response variable exceeds 2.58 (p -value < 0.01), MESF typically reduces RMSE by approximately 0.2. In contrast, under the same level of spatial autocorrelation, spatial lags yield a reduction of about 0.05. x- and y-coordinates contribute to RMSE reduction only in cases of positive spatial autocorrelation in the response variable, and the magnitude of improvement is generally smaller compared to the other spatial features.

Incorporating spatial features enhances model accuracy under second-order effects, with the degree of improvement varying across models (Figure 6). Consistent with previous findings (Figure 5), RF shows the greatest improvement across all types of spatial features. However, this outcome is likely attributable to underfitting within the simulation process. Among the remaining three models, no statistically significant differences in accuracy improvement are observed for the same type of spatial feature. Nevertheless, SVR generally achieves the highest improvement among these models, which may be due to its tendency to select a larger number of spatial features during the feature selection process (Koo et al. 2024).

Finally, this study evaluates the influence of incorporating spatial features on reducing spatial autocorrelation in model residuals, depending on the level of spatial autocorrelation in the response variable (Figure 7). The degree of residual spatial autocorrelation reduction differs notably by the type of spatial feature. In particular, incorporating MESF appears effective in mitigating spatial dependence in the residuals. Specifically,

residuals from models with MESF generally exhibit statistically insignificant spatial autocorrelation at the 5% significance level (red dotted lines in Figure 7), regardless of the spatial autocorrelation observed in the response variable.

The other two spatial features are less effective in mitigating spatial autocorrelation in the residuals. Specifically, the incorporation of x- and y-coordinates results in a residual spatial autocorrelation pattern that is largely similar to the relationship observed in Figure 4a, indicating limited impact. In contrast, incorporating spatial lags often reverses the direction of spatial autocorrelation in the residuals. When the response variable exhibits positive spatial autocorrelation, the residuals tend to show negative spatial autocorrelation, and vice versa. This inversion suggests that the spatial lag may overfit the spatial structure present in the response variable.

The reduction in residual spatial autocorrelation varies less across models than across spatial feature types (Figure 7). LM slightly differs from the other three models. Specifically, when combined with x- and y-coordinates and spatial lags, LM tends to yield higher absolute levels of residual spatial autocorrelation compared to the other models. In contrast, when MESF is incorporated, LM exhibits relatively lower levels of residual spatial autocorrelation compared to the other models. This outcome may be attributed to multicollinearity issues in LM arising from correlations between spatial lags or x- and y-coordinates. MESF, which is less susceptible to such issues, appears to provide more robust performance in mitigating spatial autocorrelation.

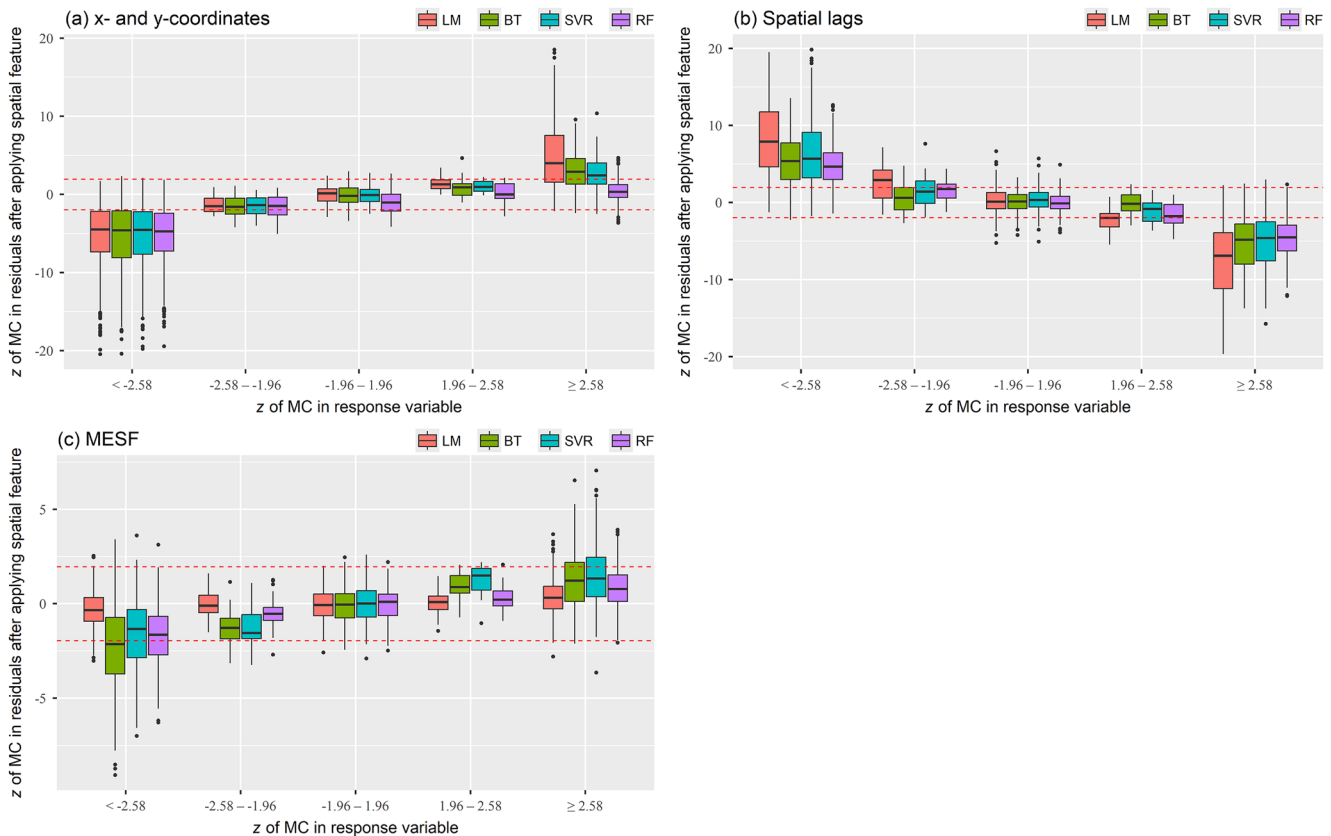


FIGURE 7 | Z-values of Moran coefficients in model residuals after incorporation of spatial features across z-values in response variables.

3.3 | Empirical Validation: Homeownership Rate

This empirical analysis aims to validate the findings of the simulation experiments using real-world data characterized by strong spatial dependence. Similar to the simulation, the evaluation focuses on two aspects: prediction accuracy, measured by the reduction in RMSE, and the effectiveness of spatial features in mitigating residual spatial autocorrelation, quantified by the z -values of the Moran coefficient. Specifically, the analysis examines whether MESF maintains its superior performance over spatial coordinates and spatial lags in a multivariate setting and whether spatial lags exhibit the tendency to overcorrect residuals as observed in the simulation. Unlike the simulation, this analysis selects eigenvectors corresponding to eigenvalues that are at least 10% of the largest eigenvalue to adequately capture the strong positive spatial autocorrelation inherent in the homeownership rate data.

The incorporation of spatial features consistently enhances predictive accuracy across all cases, with MESF yielding the largest reduction in RMSE (Table 1). In the LM, the RMSE decreases from 4.546 (baseline) to 4.487 with coordinates and to 4.289 with spatial lags. However, MESF achieves the lowest value of 3.755, representing a substantial reduction of 0.791. A similar pattern persists across other models, where the inclusion of coordinates and spatial lags lowers RMSE by 0.333 and 0.320 for BT, 1.064 and 0.663 for SVR, and 1.214 and 0.347 for RF, whereas MESF delivers the most pronounced improvements of 0.424 for BT, 1.460 for SVR, and 1.571 for RF. These results suggest the potential benefits of spatial features in machine learning models, and particularly indicate that the MESF could lead to further improvements in predictive performance.

The analysis of model residuals further indicates that applying spatial features generally mitigates residual spatial autocorrelation, with MESF being the only approach to achieve statistical insignificance at the 5% significance level across all models (Table 2). For BT, SVR, and RF, all spatial feature types effectively reduce the spatial autocorrelation, reducing the Moran coefficient to a statistically insignificant level. However, in the LM, both coordinates ($z = 2.122$) and spatial lags ($z = -2.788$) show significant spatial autocorrelation in model residuals. In contrast, MESF reduces the z -value to -1.502 , rendering it insignificant. Furthermore, a notable trend is observed with spatial lags, which shift the residuals toward negative autocorrelation in several models, specifically in LM ($z = -2.788$), BT ($z = -1.527$), and RF ($z = -0.185$). This confirms the simulation

TABLE 1 | RMSE results and differences compared to models without spatial features.

Model	Spatial feature			MESF
	Without	x- and y-coordinates	Spatial lags	
LM	4.546	4.487	4.289	3.755
BT	4.402	4.068	4.081	3.978
SVR	5.034	3.971	4.371	3.574
RF	4.472	3.259	4.125	2.901

TABLE 2 | Z -values of Moran coefficients in model residuals.

Model	Spatial feature			MESF
	Without	x- and y-coordinates	Spatial lags	
LM	2.600	2.122	-2.788	-1.502
BT	2.553	0.675	-1.527	1.858
SVR	6.447	1.644	1.522	0.568
RF	2.252	1.192	-0.185	1.906

finding that spatial lags tend to overcorrect compared to other spatial features.

4 | Conclusion

This study evaluates the effectiveness of different types of spatial features and machine learning models. The analysis is conducted through simulation experiments involving a single input variable and reaffirmed by an empirical analysis. The results indicate that the utility of spatial features varies significantly depending on the data characteristics: their impact on model accuracy is most pronounced when the correlation between input and response variables is weak (i.e., first-order effects) and when the response variable exhibits strong spatial autocorrelation (i.e., second-order effects).

Regarding the comparative effectiveness of spatial features, MESF consistently demonstrates superior performance across both aspects. It not only yields the most improvement in prediction accuracy but also effectively mitigates residual spatial autocorrelation. In contrast, while spatial lags contribute to accuracy, they tend to overcorrect, often reversing the direction of autocorrelation in the residuals.

Conversely, performance disparities attributable to model architectures remain marginal. Whether employing LM or nonlinear models (e.g., SVR, Boosting), the improvements in both accuracy and residual mitigation are generally consistent. The RF model is an exception, showing notable gains primarily because spatial features resolve its inherent underfitting in single-variable baselines. Consequently, this study concludes that the variation in effectiveness is driven more significantly by the choice of spatial features than by the machine learning algorithm itself, highlighting that identifying appropriate spatial representations is critical for both maximizing predictive power and ensuring statistical validity.

This study provides the following contributions to the field of spatial machine learning. First, through simulation, it empirically demonstrates that the incorporation of spatial features can improve the predictive performance of machine learning models and that the effectiveness of these features varies across their types. Given the growing interest in spatial machine learning and the increasing emphasis on incorporating spatial structures into models (Du et al. 2020; Jemeljanova et al. 2024; Liu et al. 2022), these findings are particularly timely and relevant. Second, the study offers a comprehensive

framework for constructing and utilizing different types of spatial features, and demonstrates their effective integration with model-specific feature selection procedures. This methodological approach not only enhances the applicability of spatial features in machine learning but also provides a practical foundation for future research, particularly in analytical contexts where spatial structure is a key consideration (e.g., Fauzi 2024; Ribeiro 2023).

This study offers the following directions for future research. First, examining the influence of varying study area sizes and shapes on the performance of spatial features would be beneficial. Although the current simulations are based on a fixed 20-by-20 grid and a single queen contiguity weights matrix, the definition of spatial relationships plays a pivotal role in capturing second-order effects. Therefore, future research should extend this evaluation to diverse spatial configurations and alternative weighting schemes to assess the sensitivity and robustness of spatial features. Second, the effectiveness of spatial features across a broader range of machine learning models warrants further investigation. While this study focuses on widely used standard machine learning models, the inherent differences in model architectures may lead to varying outcomes. Future research could extend this evaluation to neural network-based approaches, assessing the interaction of spatial features with more complex learning frameworks. In addition, while this study evaluated spatial features independently to isolate their specific contributions and minimize redundancy, future research can explore the combined use of different feature types (e.g., integrating coordinates with MESF). Such combinations may offer synergistic benefits for predictive accuracy, provided that issues of multicollinearity and model complexity are carefully managed.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A

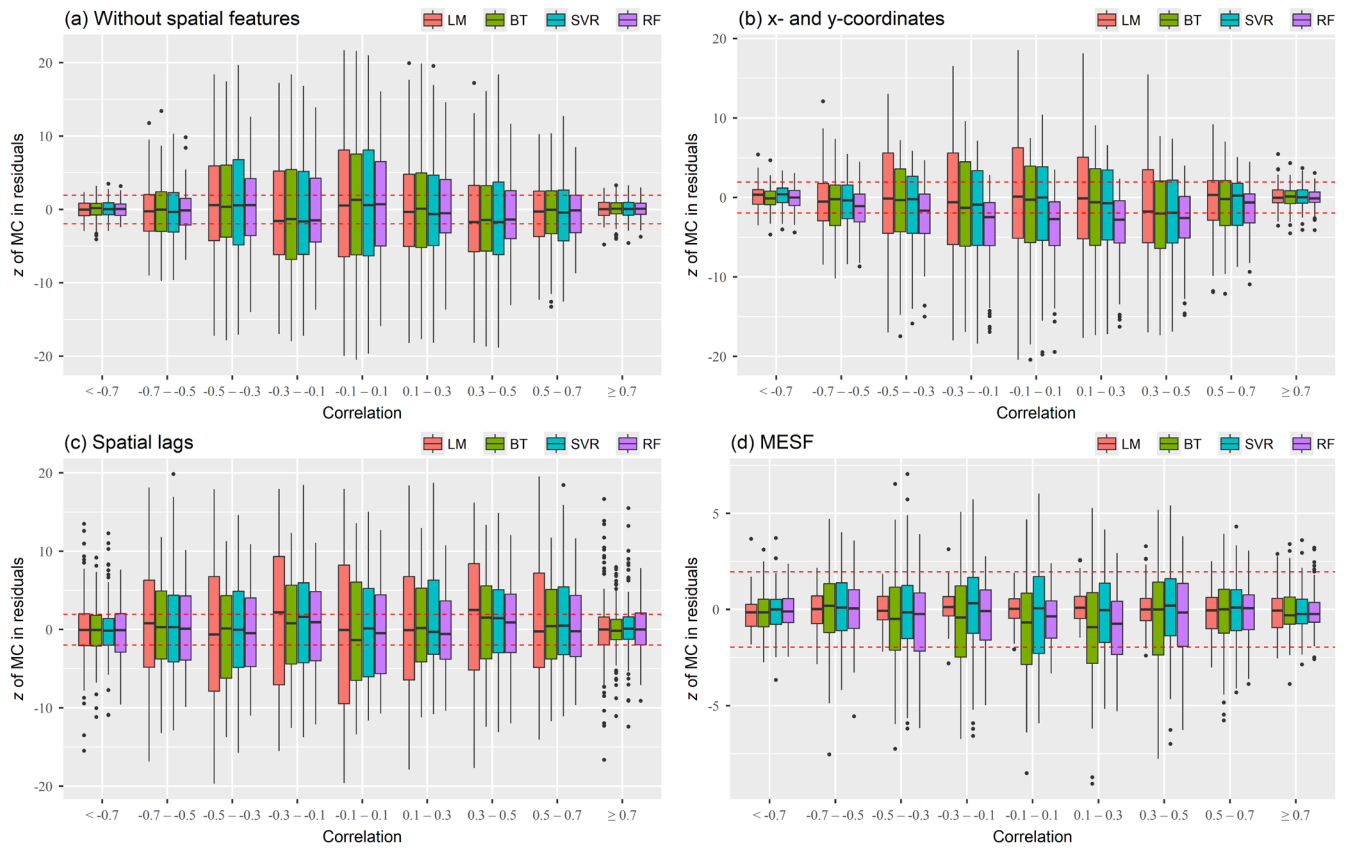


FIGURE A1 | Z-value of Moran coefficients in model residuals across correlation levels between input and response variables.